

MINI PROJECT REPORT

TITLE:

Inventory Management System for Small Business

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LIST OF ABBREVIATIONS:

- OOP - Object-Oriented Programming
- I/O - Input / Output
- IMS - Inventory Management System

INTRODUCTION:

Due to limited resources and manual processes, small businesses often need help managing inventory. A well-designed Inventory Management System (IMS) in C++ can streamline inventory tracking, reduce errors, and improve efficiency by automating stock management. This project uses object-oriented principles to create a basic C++ console-based IMS for small businesses.

PROBLEM STATEMENT:

Small businesses often face challenges in effectively managing their inventory, leading to stockouts, overstocking, manual errors, and lost sales opportunities. These inefficiencies arise from limited resources, lack of automation, and the difficulty of tracking inventory levels, sales trends, and supplier relationships manually. An automated inventory management system can help small businesses streamline operations, reduce errors, and make data-driven decisions to optimize stock levels and improve sales performance.

Feature Used:

1. Barcode Scanner Integration (Simulated):

- Simulates the functionality of a bar code scanner for quick product identification and checkout. This reduces manual input errors and speeds up the sales process.
- How it works:
 - Each product is assigned a unique barcode(which can be represented as a product ID).
 - During sales,the user can "scan" (input) the barcode to quickly add the product to the transaction.
 - The system fetches the product details based on the barcode and adds it to the sales list.

2. Stock Level Monitoring and Notifications:

- Tracks the stock levels of each product in the inventory and automatically notifies the user when stock levels fall below a predefined threshold, indicating the need for restocking.
- How it works:
 - Each product has a stock attribute that keeps track of the available quantity.
 - The system continuously monitors this attribute and checks it whenever a sale is made or stock is updated.
 - If the stock falls below a minimum threshold(e.g.,5units), the system generates a notification alerting the user to reorder the product.

3. Product Management:

- Allows the user to add new products to the inventory, update existing product details (such as price, stock, and description), and delete products that are no longer needed.
- How it works:
 - The user can input details like product name, ID, price, and initial stock quantity.
 - The system creates a new Product object and adds it to the Inventory list.
 - Users can update a product's details by searching for its productID and modifying the desired attributes.
 - The system allows deleting products by specifying the productID.

4. Sales and Inventory Reporting:

- Generates and displays different types of reports such as daily, weekly, or monthly sales reports, and inventory status reports. Provides insights into sales performance and stock levels. How it works:
 - - The user can select the type of report they want to generate(e.g.,salesfor a specific date range or a current stock report).
 - The system processes the relevant data from the Sales Report and Inventory classes to compile the report.
 - The report includes details such as totalsalesrevenue, mostsoldproducts, and current inventory status.
 - The generated report is displayed to the user, providing insights for decision-making.

LIST OF OBJECT-ORIENTED CONCEPTS USED:

- Classes and Objects
- Encapsulation
- Abstraction
- Inheritance
- Polymorphism
- Constructors and Destruction
- Member Functions and Methods
- Dynamic Memory Management
- Standard Input/Output (I/O)

OUTPUT:

Output

/tmp/5CU0oWPULI.o

Welcome to Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 1

Enter Product ID: 10

Enter Product Name: Pen

Enter Cost Price: 15

Enter Selling Price: 20

Enter Product Stock: 30

Output

Welcome to Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 1

Enter Product ID: 11

Enter Product Name: Pencil

Enter Cost Price: 5

Enter Selling Price: 10

Enter Product Stock: 45

Welcome to Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 1

Enter Product ID: 12

Enter Product Name: Ruler

Enter Cost Price: 7

Enter Selling Price: 13

Enter Product Stock: 40

Welcome to Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 1

Enter Product ID: 13

Enter Product Name: Eraser

Enter Cost Price: 5

Enter Selling Price: 10

Enter Product Stock: 35

Welcome to Inventory Management System

-
1. Add Product
 2. Update Product
 3. Delete Product
 4. Show Products
 5. Restock Notification
 6. Simulate Barcode Scanner
 7. Generate Sales Report
 8. Generate Net Profit/Loss Report
 9. Exit

Enter your choice: 1

Enter Product ID: 14

Enter Product Name: Chalk

Enter Cost Price: 50

Enter Selling Price: 65

Enter Product Stock: 100

Welcome to Inventory Management System

-
1. Add Product
 2. Update Product
 3. Delete Product
 4. Show Products
 5. Restock Notification
 6. Simulate Barcode Scanner
 7. Generate Sales Report
 8. Generate Net Profit/Loss Report
 9. Exit

Enter your choice: 4

ID: 10, Name: Pen, Cost Price: 15, Selling Price: 20, Stock: 30

ID: 11, Name: Pencil, Cost Price: 5, Selling Price: 10, Stock: 45

ID: 12, Name: Ruler, Cost Price: 7, Selling Price: 13, Stock: 40

ID: 13, Name: Eraser, Cost Price: 5, Selling Price: 10, Stock: 35

ID: 14, Name: Chalk, Cost Price: 50, Selling Price: 65, Stock: 100

Welcome to Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 3

Enter Product ID to delete: 12

Welcome to Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 4

ID: 10, Name: Pen, Cost Price: 15, Selling Price: 20, Stock: 30

ID: 11, Name: Pencil, Cost Price: 5, Selling Price: 10, Stock: 45

ID: 13, Name: Eraser, Cost Price: 5, Selling Price: 10, Stock: 35

ID: 14, Name: Chalk, Cost Price: 50, Selling Price: 65, Stock: 100

Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 2

Enter Product ID: 13

Enter New Product Name: Eraser

Enter New Cost Price: 5

Enter New Selling Price: 12

Enter New Product Stock: 25

Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 6

Enter Product Barcode: 11

Enter Quantity: 36

Sale recorded for Pencil, Quantity: 36

Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 4

ID: 10, Name: Pen, Cost Price: 15, Selling Price: 20, Stock: 30

ID: 11, Name: Pencil, Cost Price: 5, Selling Price: 10, Stock: 9

ID: 13, Name: Eraser, Cost Price: 5, Selling Price: 12, Stock: 25

ID: 14, Name: Chalk, Cost Price: 50, Selling Price: 65, Stock: 100

Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 6

Enter Product Barcode: 11

Enter Quantity: 15

Insufficient stock for Pencil

Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 7

Sales Report:

Date: Fri Oct 18 05:42:30 2024

Total Sales: 360

Inventory Management System

1. Add Product
2. Update Product
3. Delete Product
4. Show Products
5. Restock Notification
6. Simulate Barcode Scanner
7. Generate Sales Report
8. Generate Net Profit/Loss Report
9. Exit

Enter your choice: 8

Net Profit/Loss: Profit: 180

CONCLUSION:

By using these C++ and OOP concepts, a well-structured, maintainable, and scalable inventory management system will be created that effectively handles the core operations of adding, updating, deleting products, tracking sales, and generating reports. This will also provide a great learning experience in applying OOP principles to solve real-world problems.

REFERENCES:

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- 2) <https://studentprojectguide.com/c-and-c/inventory-management-system-using-c/>